

Git Documentation & Organization

Goal: The goal of the document is to provide information on the files available in Github, as well as those we could not make publicly available, so that a user could understand what information they would need to compile to run the model or apply it elsewhere.

Data Availability: This repository includes all of the novel code and public data developed for this work. However, there are two proprietary third-party datasets from the City of Santa Cruz that are available to the authors under a data use agreement but cannot be made public due to privacy and security concerns: household water billing data and some details (e.g., nodes and edges) from the City's water supply model.

Notes: We detail below which files are not able to be publicly shared, and we detail the necessary data columns or other relevant information to recreate them.

Subfolder: data

costs

- Name: Cost_SCWSM_LTVA.csv
 - Purpose: Csv file with cost values for deployment order in linear programming

data_system

- Name: Stage_Storage_Area_MG_2020_bathymetry.csv
 - Purpose: Csv file with bathymetry information for Loch Lomond

dcc_data

- Name: DCCcoefs.csv
 - Purpose: Csv file with DCC model coefficients
- Name: household_income_data.csv
 - Purpose: Csv file with single-family households data for use in simulation model
 - Columns: account, pipe size (in), pool, bathrooms, main area (sqft), tax value, census tract/block group, map_inc_1 ... map_inc_10, mean residuals (real)
 - Note: file not included in Github repository

Input_climate_stress_test.zip

- Note: This data is too large for Github and so it is directly linked to Zenodo.
- Subfolder: FLOW
 - Purpose: Csv files with the streamflow data for each stochastic simulation

- Columns: Date; Streams (raw): Bigtrees, Laguna, Liddel, Majors, Newell, Taitside; Streams (Na): Bigtrees, Laguna, Liddel, Majors, Taitside; is it a critically dry year? (binary); first flush (binary)
- Subfolder: WEATHER
 - Purpose: Csv files with the weather data for each stochastic simulation
 - Columns: Date, Precipitation, Evaporation, Turbidity (San Lorenzo River), Turbidity (North Coast), Turbidity (North Coast Majors)
 - Note: Turbidity is binary

climate_monthly.zip

- Contains: files for each stochastic climate scenario with monthly precipitation, temperature, and evapotranspiration data for us in the econometric model

Subfolder: model_assumptions_and_scenarios

- Name: cashflow_rate_assumptons.json
 - Purpose: JSON file with assumptions related to the cashflow model
- Name: inf_planning_assumptions.json
 - Purpose: JSON file with assumptions related to the infrastructure financing model, including infrastructure options (costs, deployment time, etc.)
- Name: model_assumptions_CSTE.csv
 - Purpose: Csv file with water resources model assumptions involving single values
 - Columns: Assumption (name of assumption), SCWSM-Option_Analysis... (version name, most are based on the demand scenario, examples include: SCWSM-Demand_Baseline)
 - Note: contains proprietary data not able to be included in Github repository
- Name: model_assumptions_PROFILES.xlsx
 - Purpose: Excel file with water resources model assumptions requiring monthly profiles
 - Columns: Assumption (name); 1, 2, ... (month of year)
 - Tabs: Names for each version, i.e., SCWSM-Option_Analysis
 - Note: contains proprietary data not able to be included in Github repository
- Name: used_assumptions_CSTE.csv:
 - Purpose: contains the single values utilized in the model
 - Note: File created in simulation setup
- Name: used_assumptions_PROFILES.csv
 - Purpose: contains the monthly profiles utilized in the model
 - Note: File created in simulation setup
- Name (example): file_SA.json
 - Purpose: JSON file with sensitivity analysis-relevant parameters

- Notes: The code has been designed to be able to input an empty name for this file or a file with any combination of the possible sensitivity analysis parameters. In this example, all filenames are the baseline files, except resampled_households.csv, which is not included for data privacy reasons.
- Name: Hydrological_Condition_Types.csv
 - Purpose: Csv file containing the values for the hydrological condition type by month of year

Subfolder: model > SCWSM-Option_Analysis > Parent

- Name: CST_SCWSM_OPTION_ANALYSIS_SA.json
 - Purpose: JSON file containing the parameters used to build the Pywr model
 - Notes: Parts of this file have been removed including all nodes, all edges, some parameters, and some recorders
- Note: create subfolder to save individual json files into for stochastic simulations

Subfolder: scripts

- Name: Setup_SCWSM_Option_Analysis_CST.py
 - Purpose: Python file that sets up the Pywr model for that specific simulation, including importing the climate data and the sensitivity analysis parameters
- Name: household_demands.py
 - Purpose: Python file that defines the class householdDemand
- Name: scwsim_demands.py
 - Purpose: Python file that contains all of the parameter classes input into Pywr built for this project and paper
- Name: scwsim_alternative_supply_option_operations.py
 - Purpose: Python file with the parameter classes for the alternative water supply options and operations
 - Parameter classes for specific water resources system removed, including:
 - SUPPLY_GAP: specific to SCWD
 - FELTON_DIV_TO_LL: specific to SCWD
 - DESALINATION_PLANT_YIELD/DESALINATION_PLANT_YIELD_ROF,: determines the desalination plant yield based on the month of the year and storage levels
 - DIRECT_POTABLE_REUSE_PLANT_YIELD: determines the DPR plant yield based on the month of the year and storage levels
 - DEMAND_AFTER_RATIONING: determines the value of reduced urban demand depending on if rationing/curtailment is implemented in the model
 - TRANSFER_YIELD: returns the transfer volume from x neighboring water district based on reservoir levels
 - MCGWB_SWITCH: a switch that turns the capacity of extraction to 0 on May 1st based on reservoir storage

- Soquel_SWITCH: a switch that turns the supply from Soquel ASR on/off based on reservoir levels- checked between May and September
- Name: scwsm_model_assumptions.py
 - Purpose: Python file that imports assumptions from relevant csv files
 - Note: Csv files imported not included in Github repository
- Name: scwsm_operations_and_constraints.py
 - Purpose: Python file that includes parameter classes for relevant operations and constraints
 - Parameter classes:
 - Max_flow_below_LL: sets the maximum flows downstream of the reservoir based on the month of the year
 - FELTON_DIV_TO_LL: calculates the flow pumped from Felton to the reservoir
 - Hydraulic_constraint: calculates the hydraulic loss factor for a pump in the system
 - Note: not included in Github repository
- Name: scwsm_water_rights.py
 - Purpose: Python file that includes parameter classes for relevant water rights assumptions and operations
 - Parameter classes:
 - Felton_to_LL_Annual_License: calculates the annual license at a diversion within the system
 - WATER_RIGHT_NEWELL_TO_GHWTP: returns current value of reservoir license tied to use of upstream source for water supply
 - WATER_RIGHT_NEWELL_DIV_TO_LL: returns current value of license tied to certain diversion
 - PRECIP_WATER_IN_LL: returns the volume of water from local rainfall over the reservoir
 - FELTON_WATER_IN_LL: returns the volume of water in the reservoir originating from a certain source
 - NEWELL_WATER_IN_LL: calculates the volume of water in the reservoir from a specific upstream source
 - USEABLE_PRCP_TO_GHWTP: returns volume available from precipitation in reservoir available for water treatment plant
 - WATER_MONTH_TYPE: returns the water month type based on flow from an upstream source
 - SWITCH_MCASR_INJECTION: returns the binary value based on the water month type parameter
 - SWITCH_TRANSFERS returns the binary value based on the water month type parameter
 - Note: not included in Github repository
- Name: Run_simulations_paper_Sherlock.py
 - Purpose: Python file that runs the simulations used in the paper

- Note: run on HPC system; you can run it on a local machine, but can comment lines to run different sets of policy simulations. On a local machine, it will take 1-2 minutes per simulation.
- Name: Run_simulations_Sherlock_InfRates_SA.py
 - Purpose: Python file that runs the infrastructure financing & rate design sensitivity analysis used in the SI
 - Note: run on HPC system
- Name: Run_simulations_Sherlock_DemandDemographics_SA.py
 - Purpose: Python file that runs the demand & demographics sensitivity analysis used in the SI
 - Note: run on HPC system
- Subfolder: plots
 - Name: Figure1_ModelFramework.ipynb
 - Purpose: Jupyter notebook file to create paper plot Figure 1
 - Name: Figure2_AddClimate_Sherlock.py
 - Purpose: Python file to process the data for and create paper plot Figure 2
 - Note: this figure involves a lot of data and should be run on an HPC system
 - Name: Figure3_AddClimate_Sherlock.py
 - Purpose: Python file to create paper plot Figure 3
 - Note: this figure involves a lot of data and should be run on an HPC system
 - Name: Figure4_Sherlock.py
 - Purpose: Python file to process and plot Figure 4
 - Note: this file also contains the script for plotting Figure S8
 - Name: Figure5_Sherlock.py
 - Purpose: Python file to process and plot Figure 5
 - Name: processing_functions.py
 - Purpose: Python file with functions for processing the different types of data across results figure files
 - Name: SI_FigureS2_Timeseries_Infrastructure_Deployment.ipynb
 - Purpose: Jupyter notebook file to plot the timeseries and histograms for Figure S2
 - Name: SI_FigS3_FigS4_SA_opt_results.ipynb
 - Purpose: Jupyter notebook file to plot the scatter plots and hypervolumes for the sensitivity analysis results for the optimization (Figures S3, S4)
 - Name: SI_SensitivityAnalysis_InfRates.ipynb
 - Purpose: Jupyter notebook file to create boxplots for infrastructure financing and rate design sensitivity analysis (Figure S5)
 - Name: SI_SensitivityAnalysis_DemandDemographics.ipynb
 - Purpose: Jupyter notebook file to create boxplots for demands and demographics sensitivity analysis (Figure S6)
 - Name: SI_FigS7_FigS11_policies_and_boxplots.ipynb

- Purpose: Jupyter notebook file to create the boxplots and histograms for Figures S7, S11
- Name: SI_FigureS9_baseline_infrastructure_policy.ipynb
 - Purpose: Jupyter notebook file to create the timeseries in Figure S9
- Name: SI_FigureS10_number_of_inf_options.ipynb
 - Purpose: Jupyter notebook file to create the step plots in Figure S10 comparing infrastructure planning approaches
- Name: SI_FigS14_FigS15_ROF_Dynamics.ipynb
 - Purpose: Jupyter notebook file to illustrate the ROF dynamics shown in Figures S14, S15
- Name: SI_FigureS16_hypervolume.ipynb
 - Purpose: Jupyter notebook file to plot the hypervolume for the baseline optimization results
- Name: SI_FigureS17_Data_Fit_Income_Group.R
 - Purpose: R file to plot the correlation between measured and modeled water demands across income groups (Figure S17)
- Name: SI_FigureS18_Fig2_IncomeEstimation_Sherlock.py
 - Purpose: Python file to plot a different income estimate for Figure 2 (Figure S18)
- Name: SI_FigureS19_Fig3_IncomeEstimation_Sherlock.py
 - Purpose: Python file to plot a different income estimate for Figure 3 (Figure S19)
- Name(s): SI_FigureS20_Fig4_AR_IE_Sherlock.py,
SI_FigureS20_ProcessData_AR_IncomeEstimate_Sherlock.py
 - Purpose: Python files (2) to process and plot a different income estimate for Figure 4 (Figure S20)
- Name: SI_FigureS21_Fig5_IncomeEstimation_Sherlock.py
 - Purpose: Python file to plot a different income estimate for Figure 5 (Figure S21)
- Name: SI_FigureS23_Tiimeseries_Revenue_Rate_Assumptions.ipynb
 - Purpose: Jupyter notebook file to illustrate the impacts of building infrastructure on water rates within the model (Figure S23)
- Name: SI_TableS8_income_estimates_Sherlock.py
 - Purpose: Python file to compile the statistics for the income estimates in Table S8
- Subfolder: SI_CMIP6
 - Name: cal_adapt_prec.csv
 - Purpose: CalAdapt precipitation data
 - Name: cal_adapt_temp.csv
 - Purpose: CalAdapt temperature data
 - Name: cmip6_prec.csv
 - Purpose: CMIP6 precipitation data
 - Name: cmip6_temp.csv
 - Purpose: CMIP6 temperature data

- Name: plot_CMIP6_figure.py
 - Purpose: Python file for plotting climate anomalies (Figure S1)

Subfolder: results

- Name: df_results_P100T0_dCV1.0_real1270_demandBaseline.csv
 - Purpose: Csv file containing daily water system simulations results
- Name: df_cashflow_P100T0_dCV1.0_real1270_demandBaseline.csv
 - Purpose: Csv file containing monthly cashflow model simulation results
- Name: df_time_tracker_P100T0_dCV1.0_real1270_demandBaseline.csv
 - Purpose: Csv file containing the infrastructure options deployed, their costs, and timing.
- Note: These are sample water system results for a single realization are included in the repository. We have only included results files for data that is not aggregated at the household level.